

Solution Brief

Structured Deadline Extraction and Calendar Integration for Moodle

1 Executive Summary

University students must manage academic deadlines across multiple, fragmented communication channels. While some dates appear in the Moodle calendar, a significant portion, estimated at 50%, are embedded exclusively in unstructured formats such as PDF syllabuses, assignment descriptions, SharePoint documents, and HTML-based exam systems. Consequently, important deadlines remain partially invisible, leading to missed submissions and administrative overhead.

Syllabus-Sync bridges this gap by leveraging Generative AI to convert unstructured course information into structured, actionable calendar events. Students can upload course documents or paste raw text to automatically identify and normalize relevant deadlines. These structured events are then reviewed and exported directly into personal calendars. The solution operates with a privacy-conscious, stateless architecture, ensuring maximum user control without requiring system-level access to Moodle or personal email accounts.

1.1 Business Value

The implementation of Syllabus-Sync delivers immediate, quantifiable value to both the student body and the academic administration:

- **Enhanced Student Experience:** By consolidating fragmented communication channels into a single, consistent output format, students gain improved transparency and planning reliability. This reduces the administrative friction of manually monitoring multiple systems.
- **Operational Efficiency:** Structured calendar entries directly lower the error rate of missed or late submissions. For academic staff and administrative units, this translates to fewer deadline-related queries and repetitive clarification requests.
- **Strategic Impact:** With future user consent, structured deadline data could be forwarded to departmental secretariats to streamline academic planning. Ultimately, avoiding even a single missed submission per student per semester yields a high return on investment for the university's operational efficiency.

1.2 Use Case Description

Target Audience:

- Primary: University students seeking planning reliability.
- Secondary: Academic staff and administrative units benefiting from reduced late submissions.

The "As-Is" Process:

Currently, academic planning sources are highly decentralized. Deadlines are distributed across Moodle assignments, PDF syllabuses, HTML pages, SharePoint folders, and room reservation tools. Students are forced to manually consolidate this information and create their own calendar entries. Because deadlines mentioned exclusively in documents do not automatically populate the Moodle calendar, the manual process is inefficient, time-consuming, and highly error-prone.

The "To-Be" Process:

1. The user uploads relevant course documents (PDF, Word, Excel) or pastes raw text into the application.
2. The system's Generative AI identifies, categorizes, and standardizes deadlines and task types.
3. The user reviews and validates the extracted events via a human-in-the-loop interface.
4. The validated calendar entries are exported in a standardized .ics format.
5. The user imports the .ics file into their preferred personal calendar, complete with automated follow-up reminders.

1.3 Functional scope

Must-have:

- Document upload (PDF, Word, HTML, Excel) and direct raw text input.
- AI-driven deadline identification (date, time, task, description).
- Multi-language support (German, English, French, Italian, Spanish).
- Date format normalization (e.g., resolving “12th January 2026” and “12.01.2026” to a standard format).
- Task categorization (exam, assignment, presentation).
- iCal (.ics) export with automated follow-up reminders.
- Time zone handling and mandatory user confirmation before export.

Could-Have:

- iCal (.ics) import capabilities.
- Extraction of course sections, rooms, and lecture times.
- Automated synchronization and direct calendar API integration.
- Holiday and block week detection.
- Extended “AI tutor” functionality.

Will Not Have (Out of Scope):

- Direct login integration with Moodle or email providers.
- Persistent user databases (the system uses temporary job storage only).
- Unlimited historical data retention or automatic background data access.
- Full write access to external calendar systems.

1.4 Feasibility Assessment

The proposed solution is highly feasible across technical, operational, and organizational dimensions. Technically, utilizing Generative AI for text interpretation is far more reliable than purely rule-based or regex approaches, which struggle with multilingual content, implicit descriptions, and inconsistent formatting. Operationally, the inclusion of a mandatory human-in-the-loop review step mitigates the risk of AI hallucinations and ensures high-quality output. Organizationally, the application is designed to be entirely decoupled; it does not interfere with existing Moodle or SharePoint infrastructure, bypassing complex integration approvals and security bottlenecks.

1.5 Data requirements

The system processes unstructured academic materials (PDF, HTML, Word, Excel, and raw text) sourced directly from user uploads. Because these documents often feature inconsistent formatting and multilingual content, the data is processed temporarily in a stateless environment. Ownership of the documents remains strictly with the user, ensuring compliance with data privacy standards.

1.6 Architecture Proposal

High-Level Workflow:

File Upload / Text Input → Text Extraction → AI Deadline Identification → Event Structuring → User Review → Calendar Export (.ics)

Design Characteristics & Technologies:

- **Stateless Processing:** No persistent database; utilizes temporary job storage (e.g., Redis) for asynchronous processing.
- **Extraction & Parsing:** Standardized document parsing libraries for PDF, Word, HTML, and Excel.
- **Intelligence:** Generative AI models for contextual deadline identification and text interpretation.
- **Standardization:** JSON schemas for internal event representation and the RFC 5545 iCalendar (.ics) standard for cross-platform export.

1.7 Implementation Roadmap

- **Initiation:** Define the exact scope, supported file formats, and JSON event structure.
- **Development:** Implement the document upload interface, AI extraction logic, validation UI, and .ics generator.
- **Deployment:** Test the prototype against a diverse set of real-world Moodle documents.
- **Optimization:** Refine AI prompts to improve extraction accuracy and expand format compatibility.

1.8 Success Criteria

- $\geq 85\%$ correct deadline identification during automated test cases.
- Less than 2 minutes required for a user to process and export deadlines per course.
- Zero automatic calendar modifications made without explicit user approval.

1.9 Risks & Mitigation

Risk	Metric	Countermeasure
Incorrect deadline identification	Error rate after user review	Mandatory human-in-the-loop confirmation step.
Ambiguous date formats	Number of flagged / unparsed cases	AI-driven standardization and manual user overrides.
Data privacy concerns	Data retention duration	Stateless processing architecture with zero persistent storage.

1.10 Open Discussion Points

- Should reminder timing (currently defaulting to 24 hours) be user-configurable in the UI?
- How should conflicting dates between multiple document sources (e.g., syllabus vs. Moodle export) be prioritized?
- Should lecturers be provided a portal to approve and forward extracted deadlines directly to departmental secretariats?
- Is the .ics export sufficient for long-term adoption, or will direct OAuth Calendar API integration (Google/Outlook) become necessary?

1.11 References

- Google Skills - Generative AI Learning Path: <https://www.skills.google/paths/118>
- Google Antigravity Documentation: <https://antigravity.google/docs/home>
- Moodle FHNW: <https://moodle.fhnw.ch/my/>
- Inside FHNW: <https://fhnw365.sharepoint.com/sites/inside-hsw-stud/>